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Measuring firms' capability to absorb external knowledge ☆

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ABSTRACT

As the economy has become increasingly knowledge-based, firms' capability to benefit from external knowledge has grown in importance. We construct a measure of this capability – absorption intensity – from information in firms' patents. To validate the measure, we exploit the American Inventors Protection Act (AIPA), which exogenously increased patent information availability. A triple-difference design shows that firms with higher absorption intensity experience greater innovation growth when more exposed to the reform. Beyond AIPA, absorption intensity predicts stronger innovation outcomes and firm growth. Our measure offers a tool to study how differences in absorption capability shape returns to external knowledge.

1. Introduction

Innovation is a primary driver of firm value and long-run performance (Kogan et al., 2017), and it often relies on ideas that originate outside the firm (Arrow, 1962; Griliches, 1979). This reliance has intensified as the U.S. economy has become increasingly knowledge-based – a shift reflected in the rising importance of intangible capital (Corrado et al., 2005, 2009; Eisfeldt and Papanikolaou, 2013; Peters and Taylor, 2017). In this environment, a firm's competitive advantage increasingly hinges on its absorptive capacity: the ability to recognize, assimilate, and exploit outside knowledge (Cohen and Levinthal, 1989). Heterogeneity in this capability shapes how effectively firms translate outside ideas into productive outcomes, including innovation and technological performance (Cohen and Levinthal, 1990; Bloom et al., 2013). Empirically, however, absorptive capacity remains difficult to measure because it reflects cognitive and organizational processes that are not directly observable.

We address this measurement challenge with a simple insight: firms with greater capacity to harness external knowledge are more likely to internalize newly disclosed ideas from other firms into their own innovation. Guided by this logic, we develop a novel measure – absorption intensity – that operationalizes this process by quantifying how frequently a firm's recently granted patents build on other firms' recently granted patents. This measure quantifies heterogeneity in firms' absorptive capabilities and underpins our analysis of firm-level interactions with innovation outcomes, external information, and long-run economic performance.

Prior research, which we review in detail later, has relied heavily on R&D-based proxies for absorptive capacity. R&D, however, provides only a noisy signal of this capability: R&D reflects both firms' investment in producing new knowledge and firms' capability to draw on external knowledge. This dual role of R&D makes it difficult to isolate absorption as a distinct mechanism (Griffith et al., 2003). Additional

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noise arises from heterogeneity in how efficiently firms convert R&D into this capability (Brown et al., 2022) and from policy incentives that distort R&D for reasons unrelated to external knowledge (Mukherjee et al., 2017). Consequently, R&D-based measures perform poorly in explaining firms' responses to information shocks. By directly capturing the extent to which a firm incorporates external information into its observable innovation output, our measure overcomes these limitations.

Our analysis begins by characterizing how absorption intensity varies across firms and industries. Firms with higher absorption intensity are smaller and younger and have fewer tangible resources. However, they generate more patents relative to their scale. The measure varies substantially across firms, even within the same industry, but it is moderately persistent within firms over time. These patterns are consistent with absorption intensity capturing a lasting, yet adaptable, component of firms' capability to absorb external knowledge.

Recognizing that absorption intensity is endogenous, we identify its causal effect on innovation by exploiting an exogenous improvement in the external information environment – the American Inventors Protection Act of 1999 (AIPA). Before AIPA, patent applications remained confidential until grant, which was often several years after filing. The reform required most U.S. patent applications to be published 18 months after filing, thereby increasing the timeliness and breadth of disclosures and sharply expanding the information available to other inventors (Hegde et al., 2023). We estimate a triple-difference design that interacts firms' pre-AIPA absorption intensity with firms' estimated exposure to the reform and a post-AIPA indicator. We find that firms with higher pre-AIPA absorption intensity experience significantly larger post-AIPA increases in innovation output when more exposed to the reform. In contrast, firms with high absorption intensity that are not exposed to the reform show no comparable increase, alleviating concerns that our results are driven by unobserved firm characteristics or contemporaneous macroeconomic shocks.

We then provide evidence consistent with the mechanism that firms with greater absorption intensity are better able to exploit newly available information after AIPA. Following the reform, these firms exhibit larger increases in citations to newly disclosed patent applications, indicating that they indeed make greater use of the expanded information set.

A possible alternative interpretation is that firms differ in their ability to scan the patent system (Sampat, 2010), and our measure captures this ability, with higher-ability firms more likely to identify and cite newly published patents. Under this view, the post-AIPA expansion in disclosure could mechanically advantage such firms, leading to increased incremental or defensive patenting rather than advances at the technological frontier. However, the evidence is inconsistent with this mechanism. Following AIPA, firms with higher absorption intensity experience significantly larger increases in breakthrough, exploratory, and non-core innovation with no corresponding rise in exploitative (incremental) innovation. Moreover, conditional on patenting, we observe no decline in patent novelty, scope, or value. These patterns indicate that the post-AIPA expansion reflects increases in frontier innovation rather than increases in incremental innovation.

Finally, we show that our findings are robust to alternative measures of absorption intensity, different R&D controls, and a wide range of citation-based and technological controls. Placebo tests using pre-AIPA years yield no significant effects, further supporting the validity of our identification strategy.

Beyond the AIPA setting, higher absorption intensity is positively associated with greater patent output, novelty, scope, and value. Firms with higher absorption intensity also experience faster growth in R&D capital, profits, and employment. These correlations suggest that absorption intensity shapes not only firms' responses to information shocks but also firms' long-run growth trajectories.

A firm's capability to absorb external ideas reflects a combination of observable and latent factors. Consistent with this view, we find that absorption intensity correlates with organizational practices that support innovation, the local research ecosystem, and measures of human capital. However, our measure relates more strongly to subsequent innovation than any of these observed factors, suggesting that absorption intensity captures the combined influence of multiple underlying forces.

Our findings help explain why innovation outcomes differ persistently across firms with similar observable characteristics (Hirshleifer et al., 2013). Prior work establishes that R&D is an incomplete proxy for innovation performance, leaving a substantial and persistent residual component of cross-firm variation. However, the mechanisms underlying this residual remain largely unexplored. We argue that at least part of this variation is associated with differences in firms' absorptive capacity. This capability reflects accumulated routines and firm-specific human capital that are not directly observable (Lev and Radhakrishnan, 2005) and is naturally interpreted as a dimension of organization capital (Eisfeldt and Papanikolaou, 2013). Therefore, our measure provides a new way to quantify an important and previously elusive dimension of firms' organization capital.

Most empirical work on firm-level innovation emphasizes incentives, constraints, and access to resources and information – such as governance (Atanasosov, 2013; Chemmanur and Tian, 2018), financial constraints (Brown et al., 2009), listing status (Bernstein, 2015), managerial incentives (Hirshleifer et al., 2012), ownership structure (Aghion et al., 2013; Li et al., 2023), and spatial networks (Giroud et al., 2026). We contribute to this literature by focusing on firms' capability to use external information, which complements standard determinants of innovation and helps explain persistent differences in innovation outcomes and long-run performance.

Our paper also contributes to the literature on firm boundaries by measuring a key determinant of the returns to acquiring external knowledge. This literature studies firm-boundary choices – such as strategic alliances (Hamel, 1991; Gomes-Casseres et al., 2006), corporate venture capital (Benson and Ziedonis, 2010; Chemmanur et al., 2014; Ma, 2020), joint ventures (Kogut, 1988; Lane et al., 2001), and mergers and acquisitions (Phillips and Zhdanov, 2013; Bena and Li, 2014; Frésard et al., 2020) – that govern access to external technology. Access alone, however, is not sufficient: the returns to these strategies depend on firms' capability to absorb external knowledge. Absorption intensity offers a practical, scalable proxy for this capability, enabling a systematic examination of cross-sectional heterogeneity in the success of information-intensive acquisitions and collaborations.

Finally, we contribute to the literature on absorptive capacity by offering a methodological advance. Existing empirical work relies on R&D-based proxies that conflate investment in knowledge creation with the capability to absorb external information.[1] In contrast, our approach identifies this capability directly from innovation output, thereby capturing how firms use external knowledge when it becomes available. This output-based perspective isolates absorptive capacity as a distinct mechanism and better explains firms' responses to information shocks.

2. Measuring absorption intensity

This section introduces and characterizes our measure of firms' capability to absorb external knowledge. We first describe its conceptual foundation and construction using patent citation data before comparing it with traditional R&D-based proxies. We then outline the data and sample used to compute the measure, examine how absorption intensity varies across firms and industries, and address potential sources of measurement error.

[1] See Zahra and George (2002), Lane et al. (2006), Volberda et al. (2010), and Song et al. (2018) for comprehensive reviews.

Table 1
Summary statistics.

	Panel A: Low absorption			Panel B: Medium absorption			Panel C: High absorption		
	Mean	Std. dev.	Obs.	Mean	Std. dev.	Obs.	Mean	Std. dev.	Obs.
Absorption intensity (unlogged)	1.73	1.01	2351	3.29	1.32	3131	6.36	3.36	2052
R&D capital	916	2724	2343	404	1503	3044	91	329	2113
Physical capital	3131	8271	2343	1676	5578	3041	674	3529	2112
Labor	15.42	36.46	2313	10.61	27.48	2998	5.27	17.46	2087
Profit	1916	4842	2343	1095	3222	3044	505	1883	2113
Patent count	3.23	2.33	2186	3.31	1.93	2915	2.13	1.44	1942
Firm age	15.6	13.7	2343	14.8	13.8	3044	11.3	10.7	2113
Tangibility	0.27	0.19	2343	0.24	0.17	3041	0.21	0.17	2113
Market-to-book	2.56	2.96	2302	2.61	2.93	2969	2.84	3.20	2092
Leverage	0.20	0.19	2339	0.18	0.17	3040	0.16	0.19	2111

This table presents summary statistics for absorption intensity and all control variables in the pre-AIPA period (1996 through 2000) for low, medium, and high absorption intensity firms. Firms are classified as low (≤30th percentile), medium (30th to 70th percentiles), and high (>70th percentile) absorption intensity based on the 2000 cross-sectional distribution. All variables are winsorized at the 1% level in each tail. See Appendix A for variable definitions.

2.1. Concept and construction

We measure firms' capability to absorb external information using patent citations, which identify prior inventions on which the focal patent builds and are widely used to trace the flow of technical information across firms (Jaffe et al., 1993, 2000). Formally, for firm i in year t,

$$Absorption\ intensity_{it} = \log\left(\frac{1}{N_{it}} \sum_{p \in \mathcal{P}_{it}} \text{Unique external cites}_{p,\tau \leq 5}\right)$$

where $\mathcal{P}_{it}$ denotes the set of patents applied for by firm i during years $t-4$ through t that were eventually granted, and N_{it} is its cardinality. $Unique\ external\ cites_{p,\tau\leq 5}$ counts the number of unique citations made by patent p to patents granted to other firms within $\tau \leq 5$ years before p's application date, where τ measures the lag (in years) between the cited patent's grant date and p's application date.[2] We take the natural log to smooth the right-skewed distribution. Intuitively, a firm with higher absorption intensity draws more heavily on recently created ideas from other firms, indicating a stronger capability to recognize and internalize external knowledge.

2.2. Comparison with R&D-based proxies

While absorption intensity directly captures how firms use external knowledge, prior work has largely inferred this capability from R&D activity. However, R&D-based measures face several limitations. First, R&D investment affects innovation through two channels: directly as the primary input to innovation output and indirectly as an input to a firm's capability to absorb and apply external knowledge. This dual role of R&D makes it difficult to isolate absorption as a distinct channel (Griffith et al., 2003). Second, firms vary widely in how effectively they convert R&D investment into the capability to absorb external information (Brown et al., 2022). This heterogeneity reflects differences in organizational features that are difficult to observe. Third, R&D investment is often influenced by policy incentives and institutional factors that are unrelated to external knowledge acquisition (Mukherjee et al., 2017), further weakening its usefulness as a proxy for absorption.

To assess the empirical distinctiveness of absorption intensity, we benchmark it against six standard R&D-based proxies: R&D capital, R&D expense, an indicator for positive R&D expense, R&D-to-assets, R&D-to-labor, and R&D-to-sales.[3] We take the natural log of all continuous proxies. In our data, the correlations between absorption intensity and these proxies range from –0.16 to 0.00. The near-zero correlations imply that innovation investment and external knowledge integration are empirically distinct: firms' R&D provides little information about the extent to which their patented inventions reflect recent external technical knowledge.

2.3. Data and sample construction

We obtain patent data from PatentsView, which is maintained by the Office of the Chief Economist of the U.S. Patent and Trademark Office (USPTO). The dataset provides information on patent identifiers, assignee names, application dates, grant dates, and citations. Patents are dated by filing date, which best reflects the timing of underlying innovations (Griliches et al., 1988). Using the patent-permco matches from Stoffman et al. (2022), we merge the patent data with firm-level financial information from CRSP/Compustat. We exclude firms in finance (SIC 60–67), utilities (SIC 490–494), and public administration (SIC 9), as well as firm-years with missing or non-positive assets or sales.

Following prior work on AIPA (e.g., Kim and Valentine, 2021; Saidi and Žaldokas, 2021; Boot and Vladimirov, 2025), we define the pre-AIPA period as 1996 through 2000. We omit 2001 and 2002 from the estimation window; AIPA had been enacted, but publications began only in mid-2002. Firms are required to have non-missing data in 2000 and at least one observation between 2003 and 2007. The final sample consists of all firm-years meeting these criteria.

Table 1 presents summary statistics for firms in the pre-AIPA sample, partitioned into low, medium, and high absorption intensity groups. Firms are assigned based on their position in the 2000 cross-sectional distribution of absorption intensity: the bottom 30% (low), middle 40% (medium), and top 30% (high). For ease of interpretation, the table reports the unlogged version of absorption intensity. Firms with high absorption intensity are systematically smaller in scale: they exhibit lower R&D capital, physical capital, labor, and profits. Firm age, asset tangibility, and leverage decline steadily with absorption intensity, while market-to-book ratios increase. Although firms with high absorption intensity have fewer patents in absolute terms, their patent counts normalized by size measures are higher.

2.4. Variation in absorption intensity

To provide context for the empirical analysis, we next examine how our measure varies across firms and industries. Fig. 1 plots the mean and dispersion of absorption intensity across the twenty three-digit SIC industries with the most observations in the pre-AIPA period, which together account for more than 60% of the sample. The measure varies considerably across industries – ranging from nearly 1.5 in Computer and Data Processing Services to just above 0.5 in Drugs – and shows substantial dispersion within industries. This within-industry variation indicates meaningful firm-level heterogeneity in absorption intensity.

We decompose variation in absorption intensity during the pre-AIPA period into between-industry, between-firm-within-industry, and

[2] We provide robustness tests using alternative windows of three and ten years.

[3] Detailed definitions for all variables are provided in Appendix A.

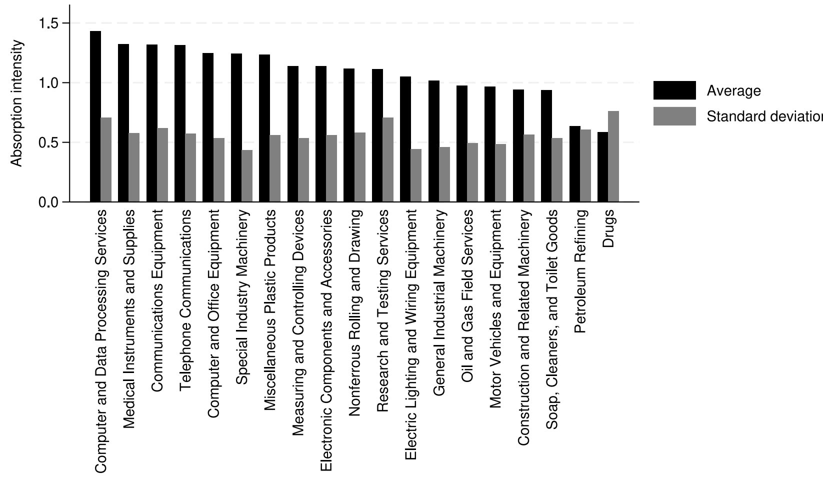

Fig. 1. Absorption intensity.
This figure presents the mean and standard deviation of absorption intensity in the twenty three-digit SIC industries with the most observations in our sample in the pre-AIPA period (1996 through 2000). Absorption intensity is winsorized at the 1% level in each tail.

within-firm components following Ma (2025). The bulk of the variation arises between firms within industries (63%), with a smaller share attributable to differences across industries (22%) and within firms (15%). A decomposition that absorbs industry-by-year effects yields similar results, which indicates that most variation occurs between firms within the same industry-year. Consistent with the limited within-firm component of this variation, absorption intensity is highly persistent, with one- and four-year autocorrelations of 0.90 and 0.56.

Between-industry variation likely reflects differences in industry-level patenting intensity rather than firm's absorption capability. In technology areas characterized by dense and cumulative innovation, such as software or electronics, the large stock of relevant prior patents makes it mechanically easier and often necessary for firms to cite patents granted to other firms (Scotchmer, 1991; Hall et al., 2001). By contrast, in industries such as pharmaceuticals, where innovation is more discrete and science-driven, firms rely less on externally patented knowledge (Henderson and Cockburn, 1996; Cohen et al., 2000). Variation in citation practices and technological conditions across industries and technological clusters, as well as over time, motivates the inclusion of industry-by-year fixed effects in the baseline specification and technological cluster-by-year fixed effects in robustness checks.

2.5. Potential measurement concerns

Before proceeding to the empirical analysis, we discuss potential sources of measurement error and bias in the construction of absorption intensity. First, absorption intensity is observable only for firms with granted patents, though this limitation parallels that associated with R&D-based proxies (Koh and Reeb, 2015).

Second, patent citations are added by both applicants and examiners during the prosecution process. Applicant-added citations more directly reflect a firm's knowledge sourcing decisions and therefore align closely with absorption activity, raising the concern that examiner-added citations may introduce noise. However, prior evidence indicates that examiners often identify relevant prior art that informed the invention but was not cited by the applicant, whether inadvertently or for strategic reasons (Hegde and Sampat, 2009; Alcácer et al., 2009). As a result, including examiner-added citations may yield a more comprehensive measure of absorption intensity. Beginning in 2002, the data distinguish citation source, which allows us in Section 4.3.3 to compare measures based on total citations, applicant-added citations, and examiner-added citations.

Third, absorption intensity could reflect strategic or stylistic citation practices rather than underlying information use. On one margin, firms may withhold prior art to preserve claim breadth, even though failure to disclose material prior art can jeopardize patent validity (Caballero and Jaffe, 1993; Cotropia et al., 2013). On another margin, firms may add external citations for reasons unrelated to absorption. For example, more incremental inventions are closer to existing technologies, mechanically increasing the likelihood of external backward citations. If such non-informational forces are important in the cross section, absorption intensity should be weakly, or even negatively, related to standard measures of patent novelty, scope, and value. However, we later show that is not the case.

Fourth, our identification leverages the AIPA-induced information shock: while AIPA may affect citation practices through multiple channels, non-informational motives would need to shift discontinuously at AIPA in a way that mimics the information mechanism to account for our estimates. Absent such a shift, these forces primarily add noise to absorption intensity and attenuate the estimated response, making the AIPA effect harder to detect.

Finally, since our measure is constructed from backward citations, it may correlate with other backward citation-based indicators of inventive behavior, such as exploratory or exploitative patents (Benner and Tushman, 2002), originality (Trajtenberg et al., 1997), technological obsolescence (Ma, 2025), or search effort (Sampat, 2010). To address this concern, we control for these proxies in robustness analyses.

3. Identification and empirical design

This section provides information on the institutional features of AIPA, outlines the empirical framework, and defines the measure of treatment intensity that captures variation in exposure to the reform across technologies.

3.1. The American Inventors Protection Act of 1999

Before AIPA, patent applications submitted to the USPTO remained confidential until grant. Technical details, drawings, and claims were publicly unavailable for several years. For example, patents filed in 2001 took an average of 38 months from application to grant (Graham and Hegde, 2015). Applications that did not result in issued patents remained secret indefinitely, preventing disclosure of innovations that never reached the grant stage.

AIPA fundamentally changed this regime. For most applications filed after November 28, 2000, AIPA requires publication 18 months after the earliest application date.[4] By reducing disclosure delays by roughly half on average and publishing previously confidential applications, the reform produced an exogenous increase in the timeliness and scope of information available to other inventors.

Building on evidence that AIPA accelerated knowledge diffusion across inventors and technologies (Kim and Valentine, 2021; Saidi and Žaldokas, 2021; Hegde et al., 2023; Boot and Vladimirov, 2025), we use the reform as a quasi-natural experiment to test whether firms better able to absorb external knowledge benefited more from this improvement in the information environment. The identifying assumption is that absent AIPA, firms with different absorption capabilities would have followed parallel trends in innovation output.

3.2. Absorption intensity and AIPA

If absorption capability determines how effectively firms transform new information into innovation, then a sudden improvement in the information environment should disproportionately benefit high-absorption firms. We test this prediction using a difference-in-differences Poisson model of patent counts and exploit AIPA as an exogenous improvement in the information environment:

$$\mathbb{E}\left[Patents_{it} \mid X_{it}\right] = \exp\big(\alpha + \beta_1 Absorption\ intensity_i \times Post_t + \eta_i + \zeta_i \times \delta_t + \gamma_i \times \delta_t\big). \tag{1}$$

$Patents_{it}$ is the number of patent applications filed by firm i in year t that are eventually granted. This variable is winsorized at the 99% level of the annual distribution among non-zero observations. $Absorption\ intensity_i$ is measured as of 2000 and winsorized at the 1% level in each tail. $Post_t$ equals zero for the pre-period (1996 through 2000) and one for the post-period (2003 through 2007). The specification includes firm fixed effects (η_i), industry-by-year fixed effects ($\zeta_i \times \delta_t$), and controls-by-year fixed effects ($\gamma_i \times \delta_t$). Industry is at the three-digit SIC level as of 2000.

The vector γ_i contains the control variables summarized in Table 1. Since the AIPA shock could affect these variables in the post period and thus bias the coefficient of interest, all controls are measured as of 2000 to avoid the "bad controls" problem (Angrist and Pischke, 2009). We interact these predetermined controls with year fixed effects, allowing their effects to vary flexibly over time. Identification therefore comes from differential changes in patenting after AIPA across firms within the same industry and year that differ in baseline absorption intensity.

To enhance comparability across firms and allow for flexible, potentially nonlinear relationships between controls and patenting outcomes, each control is discretized into quintiles. This approach reduces sensitivity to outliers and functional-form assumptions while preserving the cross-sectional ordering of firms (Bau and Matray, 2023).

3.3. Treatment intensity

The magnitude of the AIPA shock varied across technologies because disclosure acceleration depended on pre-reform grant lag (i.e., the time between a patent's application and grant date). AIPA's 18-month publication rule represented a larger relative change in disclosure timing for technologies whose patents historically took longer to issue.

We measure treatment intensity using a variable we call grant lag, which captures how strongly each firm was exposed to the disclosure acceleration introduced by AIPA. The construction mirrors the first two steps of absorption intensity, linking each firm's new patents to the external patents on which they build. Specifically, for firm i in pre-AIPA year t:

$$Grant\ lag_{it} = \frac{1}{N_{it}} \sum_{p\in\mathcal{P}_{it}} \left(\frac{1}{M_p} \sum_{q\in C_p} Lag_q \right),$$

where C_p is the set of unique external citations made by patent p to patents granted to other firms ($j \neq i$), M_p is the number of such citations, and Lag_q denotes the average application-to-grant duration of patents in the same three-digit Cooperative Patent Classification (CPC) and grant year as the cited patent q. The resulting average reflects the extent to which a firm's pre-AIPA knowledge base relied on slow-granting technologies, and therefore how much the 18-month publication rule accelerated information disclosure for that firm. Grant lag is calculated using only data prior to the introduction of AIPA, serving as a fixed, pre-treatment measure of exposure intensity.

In 2000, the mean (median) grant lag is 721 (711) days, with a minimum of 558 days (just over 18 months) and a maximum of 964 days (just over 30 months). Firms in the left tail of the distribution are therefore effectively untreated by AIPA. The overall standard deviation is 64 days. After netting out industry means, the distribution exhibits substantial within-industry variation, with a standard deviation of 45 days. A variance decomposition indicates that approximately half of the total variation in grant lag arises between industries and half within industries.

To facilitate interpretation and limit the influence of outliers, we transform grant lag into an ordinal percentile rank from 1 to 100.[5] A higher rank indicates a longer pre-AIPA delay and thus a larger acceleration in information disclosure after the reform. We then estimate a triple-difference specification to test whether firms' post-AIPA innovation responses vary with both absorption capability and exposure to faster disclosure:

$$\begin{aligned}\mathbb{E}\left[Patents_{it} \mid X_{it}\right] = \exp\big(&\alpha + \beta_1(Absorption\ intensity_i \times Post_t) \\ &+ \beta_2(Grant\ lag\ (ordinal)_i \times Post_t) \\ &+ \beta_3(Absorption\ intensity_i \times Grant\ lag\ (ordinal)_i \times Post_t) \\ &+ \eta_i + \zeta_i \times \delta_t + \gamma_i \times \delta_t\big).\end{aligned} \tag{2}$$

The coefficient of interest, β_3, captures whether firms with both (i) higher pre-AIPA absorption intensity and (ii) greater exposure to accelerated disclosure experienced larger post-AIPA increases in innovation output.

4. Results

In this section, we examine differential responses to AIPA across firms with varying absorption intensity, the underlying mechanisms, and the robustness of the results.

4.1. Absorption intensity and response to AIPA

We begin by estimating the baseline difference-in-differences specification and then examine how the effect varies with exposure to the reform.

4.1.1. Baseline effect

Panel A of Table 2 reports estimates from the baseline difference-in-differences specification (Eq. (1)). The first specification includes firm and year fixed effects and yields a coefficient of 0.78 (z-score of 6.79). Replacing year fixed effects with industry-by-year fixed effects in the second specification produces a nearly identical estimate of 0.80 (z-score of 6.59), indicating that the result is not driven by time-varying industry-level shocks. The third specification includes controls

[4] Applicants could opt out of early publication if they did not seek foreign protection and explicitly requested confidentiality, but fewer than 10% exercised this option (Graham and Hegde, 2015).

[5] Our results are qualitatively similar when using a winsorized version of grant lag.

Table 2
Baseline effect.

Panel A: Continuous absorption intensity

	5-year					3-year	10-year
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Absorption intensity × Post	0.78***	0.80***	0.64***	0.56***	0.56***	0.55***	0.69***
	(6.79)	(6.59)	(6.91)	(6.13)	(6.81)	(6.95)	(7.07)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	No	No	No	No	No	No
Industry × Year FE	No	Yes	No	Yes	Yes	Yes	Yes
Controls × Year FE	No	No	Yes	Yes	Yes	Yes	Yes
Clustering	Firm	Firm	Firm	Firm	Industry	Firm	Firm
Pseudo R^2	92.9%	93.9%	94.1%	95.1%	95.1%	95.0%	95.2%
Observations	16,798	16,158	15,293	14,615	14,615	13,156	15,906

Panel B: Absorption intensity indicators

	5-year					3-year	10-year
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Med. absorption × Post	0.38***	0.35***	0.37***	0.26***	0.26**	0.18*	0.30***
	(4.15)	(3.31)	(4.52)	(3.32)	(2.54)	(1.76)	(3.55)
High absorption × Post	0.96***	0.97***	0.77***	0.75***	0.75***	0.67***	0.78***
	(5.03)	(5.65)	(5.42)	(5.72)	(4.62)	(4.80)	(5.15)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	No	No	No	No	No	No
Industry × Year FE	No	Yes	No	Yes	Yes	Yes	Yes
Controls × Year FE	No	No	Yes	Yes	Yes	Yes	Yes
Clustering	Firm	Firm	Firm	Firm	Industry	Firm	Firm
Pseudo R^2	92.8%	93.8%	94.1%	95.1%	95.1%	95.0%	95.1%
Observations	16,798	16,158	15,293	14,615	14,615	13,156	15,906

This table presents results from Poisson regressions estimating the effect of the interaction between absorption intensity and post on innovation output (Eq. (1)). In Panel A, we use the continuous version of absorption intensity. In Panel B, we use indicator variables for medium (30th to 70th percentiles) and high (>70th percentile) absorption intensity based on the 2000 cross-sectional distribution. Controls include the variables reported in Table 1 and are binned into quintiles based on their 2000 distribution. See Appendix A for variable definitions. The 5-year, 3-year, and 10-year column headers indicate the window used to calculate absorption intensity. The sample is from 1996 through 2007 but excludes 2001 and 2002. Standard errors are clustered as indicated in that column. Statistical significance at the 10%, 5%, and 1% levels is denoted by *, **, and ***.

interacted with year fixed effects, which reduces the coefficient to 0.64 (z-score of 6.91). This attenuation suggests that these controls absorb a modest portion of the main effect and improve consistency. At the same time, statistical significance and model fit improve slightly, indicating that these controls enhance precision by explaining residual variation in outcomes. The fourth specification includes both industry-by-year and controls-by-year fixed effects, yielding a coefficient of 0.56 (z-score of 6.13).

In the first four specifications, standard errors are clustered at the firm level. As emphasized by Petersen (2009), panel regressions with persistent firm-level shocks yield severely understated standard errors if within-firm correlation is ignored, making firm-level clustering the appropriate default in our setting. We do not cluster by year because the limited number of time periods implies few time clusters, which can render cluster-robust inference unreliable.[6] As a robustness check, the fifth specification in Panel A clusters standard errors at the industry level to allow for correlated shocks across firms within industries. Industry-level clustering yields larger z-scores, implying that inference is more conservative under firm-level clustering and that within-firm dependence is quantitatively important. We therefore adopt firm-level clustering throughout the remainder of the paper.

We next assess robustness to alternative constructions of absorption intensity. Our baseline measure relies on a five-year window to capture patents and recently disclosed information. In the last two columns of Panel A, we instead use three-year and ten-year windows. The estimated coefficients are similar in magnitude and remain highly significant, indicating that our findings are not sensitive to the choice of horizon.

To facilitate interpretation of economic magnitudes, we replace the continuous absorption measure with indicator variables. Firms are classified based on their position in the 2000 cross-sectional distribution of absorption intensity: the bottom 30% (low), middle 40% (medium), and top 30% (high), with low-absorption firms serving as the omitted group.

Panel B of Table 2 shows that the corresponding estimates display a clear monotonic pattern. Relative to low-absorption firms, the differential post-period effect in our main specification (column 4) is 0.26 (z-score of 3.32) for medium-absorption firms and 0.75 (z-score of 5.72) for high-absorption firms.[7] The coefficients are economically meaningful: medium- and high-absorption firms experience approximately 30% and 112% larger expected patent counts after AIPA relative to low-absorption firms, holding firm fixed effects and other controls constant.[8]

Overall, Table 2 shows that firms with higher pre-AIPA absorption intensity exhibit markedly larger post-AIPA increases in innovation output.

4.1.2. Grant lag

We next examine whether AIPA's impact on firms with high absorption intensity varies with technological exposure to the reform. Panel A of Table 3 reports estimates from the triple-difference specification (Eq. (2)). Relative to the first specification, adding controls-by-year fixed effects in the third specification leaves the triple interaction largely unchanged but increases precision, indicating that these controls absorb residual variation rather than altering the estimated effect. As in the baseline analysis, the fourth specification includes

[6] Despite the concern, Table IA1 in the Internet Appendix shows that our results are robust to double clustering standard errors by firm and year.

[7] Although the coefficient for high-absorption firms is numerically larger than a linear extrapolation from the medium-group would predict, a Wald test fails to reject the null of linearity ($\chi^2 = 2.43$, $p = 0.12$).

[8] Although the magnitude, particularly for high-absorption firms, may appear large, it is consistent with the raw data: average patenting among high-absorption firms increases from 4.84 patents per year before AIPA to 11.21 after.

Table 3
Grant lag.

Panel A: Continuous grant lag

	(1)	(2)	(3)	(4)
Absorption intensity × Post	−0.29	−0.50*	−0.43*	−0.30
	(−1.19)	(−1.81)	(−1.96)	(−1.26)
Grant lag (ordinal) × Post	−1.50***	−1.89***	−2.22***	−2.04***
	(−3.24)	(−3.33)	(−4.98)	(−4.03)
Absorption intensity × Grant lag (ordinal) × Post	1.67***	1.94***	1.73***	1.25***
	(4.02)	(4.15)	(4.46)	(3.30)
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	No	No	No
Industry × Year FE	No	Yes	No	Yes
Controls × Year FE	No	No	Yes	Yes
Pseudo R^2	93.0%	94.1%	94.3%	95.3%
Observations	13,532	12,951	12,325	11,720

Panel B: Grant lag indicators

	(1)	(2)	(3)	(4)
Absorption intensity × Post	−0.04	−0.13	−0.19	0.01
	(−0.20)	(−0.51)	(−0.95)	(0.06)
Med. grant lag × Post	−0.67***	−0.67**	−0.91***	−0.40*
	(−2.93)	(−2.22)	(−4.13)	(−1.66)
High grant lag × Post	−1.28***	−1.44***	−1.53***	−1.08***
	(−3.98)	(−3.93)	(−5.41)	(−3.53)
Absorption intensity × Med. grant lag × Post	0.64***	0.68**	0.71***	0.37
	(2.89)	(2.34)	(3.29)	(1.57)
Absorption intensity × High grant lag × Post	1.30***	1.41***	1.22***	0.81***
	(4.36)	(4.08)	(4.52)	(2.95)
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	No	No	No
Industry × Year FE	No	Yes	No	Yes
Controls × Year FE	No	No	Yes	Yes
Pseudo R^2	93.0%	94.1%	94.3%	95.3%
Observations	13,532	12,951	12,325	11,720

This table presents results from Poisson regressions estimating the effect of the interaction between absorption intensity, treatment intensity, and post on innovation output (Eq. (2)). In Panel A, we use the continuous version of grant lag (ordinal). In Panel B, we use indicator variables for medium (30th to 70th percentiles), and high (>70th percentile) grant lag. Controls include the variables reported in Table 1 and are binned into quintiles based on their 2000 distribution. See Appendix A for variable definitions. The sample is from 1996 through 2007 but excludes 2001 and 2002. Standard errors are clustered at the firm level. Statistical significance at the 10%, 5%, and 1% levels is denoted by *, **, and ***.

both industry-by-year and controls-by-year fixed effects. Although the coefficient is modestly attenuated, it remains highly significant.

Panel A of Table 3 also shows that once treatment intensity is incorporated, the *Absorption intensity* × *Post* effect becomes small and statistically insignificant.[9] This pattern indicates that higher absorption capability translates into stronger innovation responses primarily in settings where AIPA materially expanded available information. Our results are consistent with post-AIPA innovation growth reflecting the interaction between absorption intensity and an exogenous increase in information availability, rather than unobserved firm characteristics or contemporaneous shocks.

The *Grant lag (ordinal)* × *Post* interaction is negative and significant. Since the grant lags of cited patents closely track those of firms' own patents, this coefficient captures differential exposure to the loss of pre-grant secrecy. Technologies characterized by longer grant lags previously benefited from extended confidentiality, which preserved informational advantages and supported appropriability rents. By mandating publication after 18 months, AIPA compressed this window, disproportionately reducing the private value of patenting in these technologies.[10]

Analogous to the approach for absorption intensity, Panel B replaces the continuous grant lag measure with indicator variables that classify firms into low, medium, and high grant lag groups based on the 2000 cross-sectional distribution, with the low grant lag group serving as the omitted group. The estimates reveal a monotonic increase in the triple-interaction coefficient from medium to high grant lag group. However, the coefficient is only statistically significant for the high grant lag group, with a coefficient of 0.81.[11] The coefficient implies that following AIPA, the effect of absorption intensity on the conditional mean of patenting is approximately 125% larger for firms in the high grant lag group relative to firms in the low grant lag group.

Overall, the results in Table 3 show that the impact of absorption intensity increases with exposure to AIPA. The evidence is consistent with interpreting absorption intensity as a measure of firms' capability to process and internalize newly available external knowledge.

4.1.3. Dynamics of the effect

To assess timing and identification, Fig. 2 reports estimates from a dynamic specification. We replace the post-AIPA indicator with year dummies to obtain coefficients on *Absorption intensity* × *Grant lag (ordinal)* × *Year*. The pre-AIPA coefficients are near zero and insignificant, indicating no differential innovation trends across absorption intensity

[9] The *Absorption intensity* × *Post* coefficient reflects the estimated effect for firms with a grant lag (ordinal) of zero. Although this value lies just below the observed range of 1 to 100, the extrapolation is minor: even firms in the lowest observed percentile had average pre-AIPA grant lags only 39 days longer than the post-AIPA disclosure window. A value of zero therefore approximates a firm whose pre-AIPA grant lag exceeded the new disclosure period by less than a month.

[10] Kim and Valentine (2021) show that innovation rises with rivals' grant lags but falls with a firm's own grant lag. Their analysis focuses on relative rather than absolute lags. As we show in Table IA2 of the Internet Appendix, our results are robust to controlling for their relative spillover measure.

[11] A Wald test fails to reject the null hypothesis that the coefficient scales linearly with the intensity of the disclosure shock ($\chi^2 = 0.04, p = 0.85$).

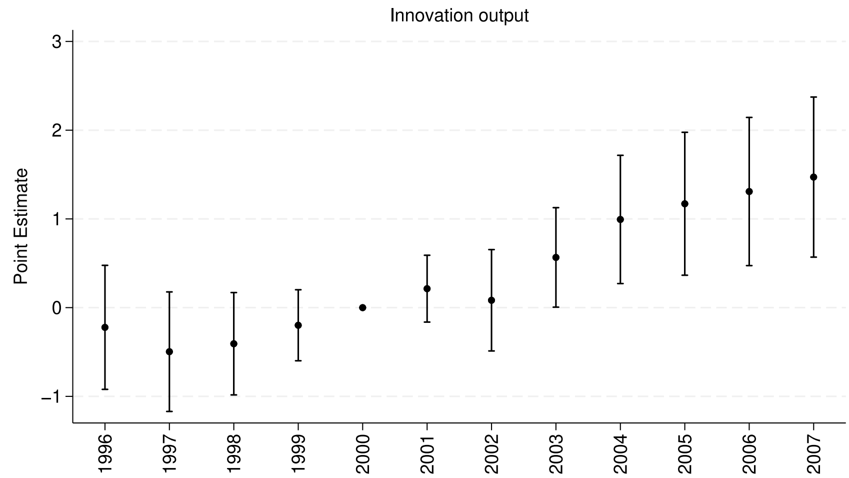

Fig. 2. Dynamics of the effect.
This figure presents results from Poisson regressions estimating the effect of the interaction between AIPA and absorption intensity on innovation output in the years around AIPA. Coefficient estimates are of the triple interaction between absorption intensity, treatment intensity, and year, which is the triple interaction from Eq. (2) after swapping out post for year indicators. Controls and fixed effects are the same as in the fourth column of Table 3. Standard errors are clustered at the firm level, and the error bars represent 95% confidence intervals.

and grant lag exposure during this period. The effect emerges in 2003, which is consistent with the timing of the first published applications under AIPA, and strengthens steadily thereafter. This pattern accords with the lag between the diffusion of newly disclosed information and its translation into patentable innovations.

In Figure IA1 in the Internet Appendix, we also assess robustness to potential deviations from parallel trends using the method of Roth (2022), which adjusts confidence intervals for the average treatment effect on the treated. The estimated effects remain statistically significant unless post-AIPA deviations exceed 50% of the largest pre-AIPA divergence ($M = 0.5$). Combined with the dynamic estimates, this robustness exercise suggests that the post-AIPA acceleration in innovation among firms with high absorption intensity and high exposure reflects a genuine causal response rather than residual trend differences.

4.1.4. Rejection rates

AIPA required quicker disclosure of applications that were ultimately granted as well as those that were rejected. Like granted applications, rejected applications reveal technical approaches, experimental designs, and problem-solving strategies, thereby expanding the stock of publicly available technical knowledge. However, the informational value of rejected patents is ambiguous ex ante. A higher rejection rate may proxy for greater disclosure volume, but it may also reflect lower average informational quality and may make information acquisition more difficult.

To examine whether firms with greater absorption intensity differentially benefit from this disclosure margin, we construct a firm-level exposure measure based on the rejection rates of the technological areas referenced in the firm's cited patents. Since rejection data are not available before AIPA, we compute rejection rates using data from 2001 through 2003, when such information becomes available. We then re-estimate our baseline specification by replacing grant lag (ordinal) with an ordinalized version of rejection rate.

Table 4 shows that the triple interaction with rejection rate is positive and significant, implying that the post-AIPA return to absorption intensity increases with exposure to rejection rates. High-absorption firms thus learn not only from granted patents but also from rejected applications. However, the coefficients are smaller and less precisely estimated than in the specification exploiting grant lag variation, indicating that this channel is weaker than the effect operating through accelerated disclosure of granted patents.

4.1.5. Heterogeneity across industries

We next examine whether the importance of absorption intensity varies across industries. In highly innovative industries, technological progress is more cumulative, and cross-firm learning is more frequent (Jaffe and Trajtenberg, 2002). These features expand the opportunities for firms to benefit from newly disclosed external information. In contrast, where innovation is less dynamic and knowledge diffusion is less important, the marginal value of absorption capability should be smaller (Bloom et al., 2013).

We assess this implication by estimating the triple-difference specification separately for highly innovative industries and other industries. We use three classifications. First, we treat high-tech industries – three-digit SIC codes 283, 357, 366, 367, 382, 384, and 737 (Brown et al., 2009) – as highly innovative. Second and third, we classify industries based on innovation intensity: industries in the top two deciles of average patent counts as of 2000 (43% of firms) and industries in the top decile of the average R&D-to-assets ratio as of 2000 (60% of firms).

Table 5 reports the results. Across all three classifications, the coefficient on the triple interaction term is positive and significant in highly innovative industries but small and insignificant in other industries. The pattern indicates that absorption capability yields the largest gains where external knowledge flows are more important.

4.2. Mechanism and alternative explanations

Having shown that firms with higher absorption intensity responded more strongly to AIPA, we next investigate the mechanism underlying this effect.

4.2.1. Use of patent application disclosures

We first examine whether firms with higher absorption intensity make greater use of the patent application disclosures made public under AIPA. We capture this behavior using citations to disclosed applications. To do so, we estimate cross-sectional regressions in which

Table 4
Rejection rates.

	(1)	(2)	(3)
Absorption intensity × Post	0.12	0.11	0.19
	(0.63)	(0.58)	(1.00)
Rejection rate (ordinal) × Post	−1.06***	−1.02***	−0.89**
	(−2.68)	(−2.65)	(−2.40)
Absorption intensity × Rejection rate (ordinal) × Post	0.63**	0.62**	0.46*
	(2.15)	(2.18)	(1.74)
Firm FE	Yes	Yes	Yes
Industry × Year FE	Yes	Yes	Yes
Controls × Year FE	Yes	Yes	Yes
Rejection rate years	2001	2001–2002	2001–2003
Pseudo R^2	95.2%	95.2%	95.2%
Observations	11,720	11,720	11,720

This table presents results from Poisson regressions estimating a modified version of Eq. (2), in which grant lag is replaced with the patent rejection rate. Rejection rates are estimated from the years indicated in that column. Controls include the variables reported in Table 1 and are binned into quintiles based on their 2000 distribution. See Appendix A for variable definitions. The sample is from 1996 through 2007 but excludes 2001 and 2002. Standard errors are clustered at the firm level. Statistical significance at the 10%, 5%, and 1% levels is denoted by *, **, and ***.

Table 5
Heterogeneity across industries.

	Hi-tech		Patent count		R&D/assets	
	(1)	(2)	(3)	(4)	(5)	(6)
Absorption intensity × Post	−0.38	0.24	−0.53	0.17	−0.06	0.03
	(−1.59)	(0.90)	(−1.52)	(0.75)	(−0.24)	(0.13)
Grant lag (ordinal) × Post	−2.36***	−0.12	−3.15***	−0.55	−1.64***	−0.87*
	(−4.09)	(−0.22)	(−3.69)	(−1.21)	(−2.76)	(−1.73)
Absorption intensity × Grant lag (ordinal) × Post	1.69***	0.07	2.08***	0.27	1.21***	0.39
	(3.78)	(0.15)	(3.39)	(0.82)	(2.68)	(0.96)
Highly innovative industry	Yes	No	Yes	No	Yes	No
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Industry × Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Controls × Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Pseudo R^2	95.6%	95.8%	96.5%	89.7%	95.7%	96.2%
Observations	6039	5603	5059	6528	6914	4706

This table presents results from Poisson regressions estimating the effect of the interaction between absorption intensity, treatment intensity, and post on innovation output (Eq. (2)) after dividing the sample into highly innovative industries and other industries. In the first two columns, highly innovative industries are industries with SIC codes 283, 357, 366, 367, 382, 384, or 737. In the middle two columns, highly innovative industries are industries in which the average patent count (as of 2000) is in the top two deciles. In the last two columns, highly innovative industries are industries in which the average R&D/assets ratio (as of 2000) is in the top decile. Controls include the variables reported in Table 1 and are binned into quintiles based on their 2000 distribution. See Appendix A for variable definitions. The sample is from 1996 through 2007 but excludes 2001 and 2002. Standard errors are clustered at the firm level. Statistical significance at the 10%, 5%, and 1% levels is denoted by *, **, and ***.

the dependent variable is the number of external application citations per patent filed between 2003 and 2005, and the main independent variable is absorption intensity. Table 6 reports the results. The first column reports Poisson estimates. The coefficient indicates that a one-unit increase in absorption intensity is associated with a 123% increase in external application citations. The second column reports ordinary least squares estimates as a robustness check, which yield similar results.

Since information on patent applications became available only after AIPA, a difference-in-differences design is infeasible. The estimates in Table 6 should therefore be interpreted as descriptive rather than causal. Nonetheless, the pattern is clear: firms with greater absorption capability were significantly more likely to incorporate newly disclosed application information into their own patents. These results are consistent with a mechanism whereby firms with higher absorption intensity more effectively process and internalize external knowledge, leading to greater productivity following AIPA.

4.2.2. Innovation inputs

An alternative explanation for our findings is that firms with higher absorption intensity responded to AIPA by expanding innovation inputs, rather than by more effectively utilizing external knowledge. If the increase in patenting reflects greater investment in innovation, we would expect corresponding increases in key inputs following the reform.

To assess this possibility, Panel A of Table 7 examines whether high-absorption firms differentially adjust innovation inputs after AIPA. Specifically, we estimate the same triple-differences specification using measures of R&D capital (Peters and Taylor, 2017), physical capital (İmrohoroğlu and Tüzel, 2014), and employment as dependent variables.

We find no evidence of changes in innovation inputs. Across all three measures, the triple interaction is economically small and statistically indistinguishable from zero. These results suggest that the increase in patenting among high-absorption firms does not reflect expanded investment in innovation inputs.

4.2.3. Patenting strategy

We next examine whether our results reflect strategic changes in patenting behavior. Alternative mechanisms based on adjustments in patenting strategies imply distinct predictions for patent novelty, scope, and value, allowing us to distinguish information absorption responses from strategic patenting responses.

One mechanism is that firms adjust how they protect and commercialize innovation following AIPA. Drawing on the trade-off between patenting and secrecy in appropriating returns to innovation (Hall et al., 2014), earlier disclosure could shift firms' reliance between formal and informal intellectual property protection. Under this mechanism, firms may split inventions into narrower patents, increase incremental or defensive filings, or pursue lower-value innovations that reduce proprietary exposure (Hegde et al., 2023; Boot and Vladimirov, 2025).

A second mechanism is that firms differ in their ability to scan the patent system (Sampat, 2010). Firms with greater scanning ability

Table 6
Use of patent application disclosures.

	(1)	(2)
Absorption intensity	0.79***	1.41***
	(4.00)	(3.86)
Controls	Yes	Yes
Industry FE	Yes	Yes
Pseudo R^2	29.7%	
Adjusted R^2		6.8%
Observations	1697	1697

This table presents results from regressions estimating the correlation between absorption intensity and the use of patent application disclosures. The dependent variable is the number of citations of external applications (made by the applicant) per patent filed in 2003 through 2005 and is winsorized at the 99% level of the non-zero observations. Controls include the variables reported in Table 1 and are binned into quintiles based on their 2000 distribution. See Appendix A for variable definitions. Standard errors are clustered at the industry level. Statistical significance at the 10%, 5%, and 1% levels is denoted by *, **, and ***.

are more likely to identify relevant prior art and cite recently granted patents. If our measure captures this ability, these firms could mechanically benefit from the post-AIPA expansion in disclosure by increasing incremental or defensive patenting without advancing the technological frontier. In this case, our estimates would reflect differential search effort rather than genuine differences in absorptive capacity.

Both mechanisms imply that any post-AIPA increase in patenting reflects a change in patent strategy, which would yield patents with narrower scope, lower novelty, and diminished value. In contrast, if the mechanism operates through knowledge absorption, high-absorption firms should produce more breakthrough patents, more exploratory patents, and expand into new technological areas. To distinguish between these interpretations, we re-estimate our triple-difference specification using measures of patent type, novelty, scope, and value as dependent variables.

Panel B of Table 7 examines changes in patenting strategy along the extensive margin. We estimate specifications in which the dependent variables are counts of breakthrough, non-breakthrough, exploratory, exploitative (incremental), core, or non-core patents. Breakthrough patents are textually distinct from prior patents but textually similar to subsequent patents (Kogan et al., 2021). Exploratory (exploitative) patents primarily cite previously uncited (previously cited) prior art (Benner and Tushman, 2002). Core (non-core) patents fall within (outside) a firm's main technological classes based on its pre-AIPA patent portfolio (Akcigit and Kerr, 2018).

The triple interaction is positive and significant for breakthrough, non-breakthrough, exploratory, core, and non-core patents, and insignificant for exploitative patents. The relative magnitudes of coefficients indicate a disproportionate expansion in breakthrough, exploratory, and non-core patenting, rather than a shift toward incremental patenting.

Panel C reports intensive-margin changes in novelty, defensiveness, and scope. Novelty is captured by originality and generality, which measure the dispersion of backward citations and forward citations across technological fields (Trajtenberg et al., 1997). Defensiveness is proxied by the share of self-citations, reflecting reliance on internal knowledge and the strategic use of citations to reinforce intellectual property boundaries (Hall et al., 2005; Alcácer and Gittelman, 2006). Scope is measured using the number and length of independent claims (Marco et al., 2019). The triple interaction is economically small and statistically insignificant across all specifications.

Panel D of Table 7 reports intensive-margin changes in proxies for patent value. Forward citations capture the extent to which a patent influences subsequent innovations (Trajtenberg, 1990). Abnormal stock returns around patent grant announcements reflect the market's assessment of economic value (Kogan et al., 2017). Renewal behavior captures private value, as more valuable patents are more likely to

Table 7
Innovation inputs and patenting strategy.

	Dependent variable	Triple interaction	z-score	Observations
Panel A: Innovation inputs				
(1)	R&D capital	0.02	(0.08)	11,720
(2)	Physical capital	−0.07	(−0.40)	11,700
(3)	Labor	0.00	(0.01)	11,575
Panel B: Patent type				
(1)	Breakthrough	2.13***	(2.82)	11,720
(2)	Non-breakthrough	1.28***	(3.55)	11,720
(3)	Exploratory	1.36***	(3.59)	11,720
(4)	Exploitative	0.52	(1.11)	11,720
(5)	Core	1.26***	(3.46)	11,720
(6)	Non-core	1.62***	(2.81)	11,720
Panel C: Patent novelty and scope				
(1)	Originality	0.09	(1.02)	6902
(2)	Generality	0.21	(1.37)	6902
(3)	Self-cites share	0.30	(1.33)	6902
(4)	Claims	0.09	(0.99)	6902
(5)	Words	−0.03	(−0.30)	6902
Panel D: Patent value				
(1)	Forward citations	−0.01	(−0.05)	6902
(2)	Economic value	0.01	(0.04)	6902
(3)	Renewal share	0.07	(0.50)	6902
(4)	Technological obsolescence	0.12	(1.12)	6784

This table presents results from regressions estimating the effect of the interaction between absorption intensity, treatment intensity, and post on innovation inputs and outputs. The dependent variable is labeled in each row. All estimates are from Poisson regressions, except when technological obsolescence is the dependent variable, in which case we use OLS. In Panel A, when an innovation input serves as the dependent variable, it is excluded from the control set. Otherwise, controls and fixed effects are the same as those in the fourth column of Table 3. See Appendix A for variable definitions. The sample is from 1996 through 2007 but excludes 2001 and 2002. Standard errors are clustered at the firm level. Statistical significance at the 10%, 5%, and 1% levels is denoted by *, **, and ***.

be maintained (Sampat, 2010). Technological obsolescence reflects the rate at which a firm's existing knowledge base loses relevance over time (Ma, 2025). The triple interaction is again economically small and statistically insignificant across all specifications.

Overall, the results in Table 7 provide little support for explanations based on a shift toward incremental, defensive, or lower-value patenting. Instead, high-absorption, highly treated firms respond to AIPA by expanding patenting – particularly in breakthrough, exploratory, and non-core technological domains – without reducing average scope or value. This pattern is most consistent with a mechanism rooted in information absorption.

4.3. Robustness

We next assess the robustness of our findings by varying the specification, conducting placebo tests with pre-AIPA years, and examining the role of citation source in measurement.

4.3.1. Alternative specifications

We start by verifying that the core triple-difference results do not depend on a particular specification or set of controls. We present the results in Table IA1 of the Internet Appendix.

In Panel A, we re-estimate our main specification using alternative R&D measures, sequentially replace the baseline R&D capital control with alternative R&D-based proxies. Across all specifications, the triple interaction term remains highly significant, indicating that the main result is not driven by how R&D intensity is measured.

In the baseline model, all controls are binned, fixed at their pre-AIPA (year 2000) levels, and interacted with year fixed effects. This

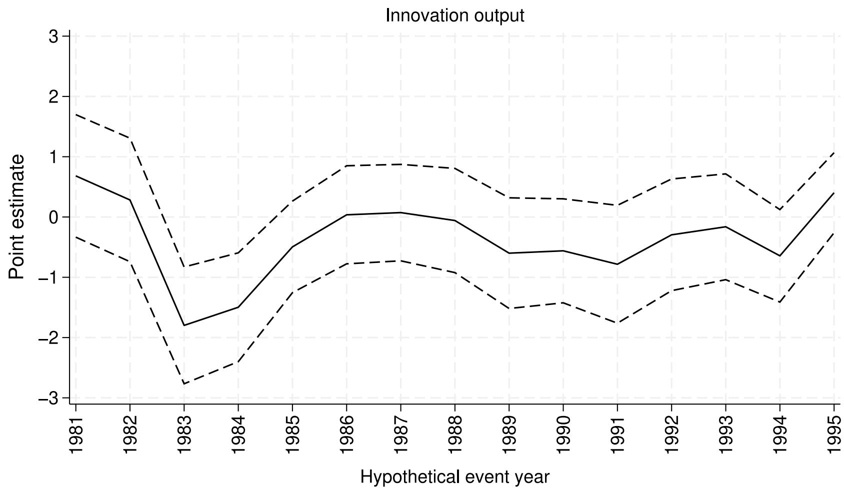

Fig. 3. Placebo tests.
This figure presents results from Poisson regressions estimating the effect of the interaction between absorption intensity, treatment intensity, and post on innovation output in the years before AIPA (Eq. (2)). To obtain these estimates, we re-estimate our main specification separately for each of 15 hypothetical event years from 1981 to 1995. Controls and fixed effects are the same as in the fourth column of Table 3. For each hypothetical event year, the sample is *t-4* through *t+7* and excludes *t+1* and *t+2*. Standard errors are clustered at the firm level, and the dashed lines represent 95% confidence intervals.

approach avoids the "bad controls" problem emphasized by Angrist and Pischke (2009). In Panel B, we relax this restriction by including one-year lagged R&D expense, capital expenditures, and labor as time-varying controls. The triple interaction coefficient declines modestly but remains economically large and statistically significant.

In Panel C, we examine whether our results are driven by other citation-based measures of inventive behavior. Specifically, we re-estimate our main specification after controlling for exploratory patents, exploitative patents, originality, technological obsolescence, examiner share of applicant self-citations, and relative spillover. In all specifications, the triple interaction term remains significant.

In Panel C, we also test whether our results reflect citation norms tied to technological rather than industrial classifications. Following Jaffe (1986) and Bloom et al. (2013), we compute technological proximity between firms based on the distribution of their patent counts across technology classes. We then apply average-linkage clustering to this measure to form 100 technological clusters. This procedure begins with each firm as its own cluster and iteratively merges clusters based on the average pairwise distance between firms. Including these technological cluster-by-year fixed effects yields slightly stronger coefficients.

Overall, the results in Table IA1 show that the estimated interaction between absorption capability, treatment intensity, and the AIPA shock is stable across alternative R&D definitions, controlling for inputs to production after AIPA, citation-based controls, and technological clustering controls.[12]

4.3.2. Placebo tests

If the estimated effects truly reflect AIPA's information shock, similar patterns should not arise in periods without a comparable disclosure reform. To verify this implication, we conduct a series of placebo analyses using pre-AIPA data.

Specifically, we re-estimate the main triple-difference specification over fifteen alternative event windows centered on hypothetical reform years between 1981 and 1995. For each placebo year, we recompute treatment exposure, redefine pre- and post-periods, and apply the same two-year exclusion window as in the baseline design.

Fig. 3 plots the estimated coefficients on *Absorption intensity* × *Grant lag (ordinal)* × *Post* for each placebo year. The solid line depicts point estimates and the dashed lines show 95% confidence intervals. None of the placebo coefficients are positive and statistically significant, confirming that the triple-difference results are not driven by persistent firm characteristics or spurious pre-trends.

4.3.3. Citation source: Applicant versus examiner

As discussed above, citations are added by both applicants and examiners during the patent prosecution process. The presence of examiner-added citations raises the question of whether absorption intensity is sensitive to citation source. Beginning in 2002, the data distinguish between applicant- and examiner-added citations. This distinction allows us to examine how measurement varies across sources.[13]

Examiner-added citations account for 24% to 29% of external backward citations among patents filed between 2003 and 2005. Despite this sizable contribution, citation counts from the two sources are highly correlated: at the firm level, the log number of examiner-added citations correlates 0.75 with the log number of applicant-added citations. Absorption intensity computed using all citations correlates between 0.84 and 0.90 with the measure based solely on applicant-added citations.

To assess the implications for measurement, we recompute absorption intensity for the post-AIPA years 2003 through 2005 using three forward-looking measures: (i) all citations, (ii) applicant-added citations only, and (iii) examiner-added citations only. We then re-estimate the main specification using each measure.[14] Although this exercise relies on post-AIPA data, it leverages the high within-firm persistence of absorption intensity.

[12] Tables IA3 and IA4 in the Internet Appendix report "horse-race" regressions in which both absorption intensity and each alternative variable from Panels A and C of Table IA1 are interacted with grant lag and post in the same specification. In every case, the triple interaction involving absorption intensity remains positive and highly significant, and the corresponding interaction for the other variable is non-positive.

[13] Even beginning in 2002, some citations labeled as examiner-added were in fact supplied by applicants (Kuhn et al., 2020), so measures based solely on applicant-added citations may still contain limited measurement error due to misclassification of citation source.

[14] For example, the 2003 measure includes applicant-added citations from patents filed between 1999 and 2003. Since citation source is identifiable only for patents granted after January 2002, this measure omits citations from patents filed before 2002 and granted earlier. Given the average grant lag of about three years, few citations are excluded. Later measures are more complete but increasingly distant from the AIPA event year.

Table 8
Citation source: Applicant versus examiner.

	All citations			Applicant-added			Examiner-added		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Absorption intensity × Post	−0.35	−0.35	−0.46**	−0.24	−0.33**	−0.34**	0.71***	0.12	−0.28
	(−1.43)	(−1.49)	(−1.96)	(−1.43)	(−1.97)	(−2.00)	(3.35)	(0.57)	(−1.13)
Grant lag (ordinal) × Post	−1.77***	−1.46**	−1.28**	−0.87**	−0.82**	−0.79**	−0.77**	−0.75**	−0.71*
	(−3.24)	(−2.61)	(−2.28)	(−2.58)	(−2.33)	(−2.16)	(−2.46)	(−2.39)	(−2.02)
Absorption intensity × Grant lag (ordinal) × Post	1.08***	0.80**	0.76*	0.87***	0.91***	0.69**	−0.20	0.29	0.56
	(2.78)	(2.06)	(1.96)	(2.84)	(2.92)	(2.40)	(−0.60)	(0.85)	(1.47)
Absorption year	2003	2004	2005	2003	2004	2005	2003	2004	2005
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry × Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls × Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Pseudo R^2	95.09%	95.24%	95.28%	95.27%	95.19%	95.32%	95.27%	95.22%	95.27%
Observations	10,700	10,060	9456	9859	9675	9080	10,070	9688	9144

This table presents results from Poisson regressions estimating the effect of the interaction between alternative versions of absorption intensity, treatment intensity, and post on innovation output (Eq. (2)). In the first three columns, we use future versions of absorption intensity that include all citations. In the middle three columns, we use an applicant-added citations version of absorption intensity. In the last three columns, we use an examiner-added citations version of absorption intensity. Each version is constructed exactly as the main version except that we only use citations added by the applicant (in the middle three columns) or by the examiner (in the last three columns). As indicated by the "Absorption year" row, each version of absorption intensity used in these tests is constructed using information as of 2003, 2004, or 2005. Controls include the variables reported in Table 1 and are binned into quintiles based on their 2000 distribution. See Appendix A for variable definitions. The sample is from 1996 through 2007 but excludes 2001 and 2002. Standard errors are clustered at the firm level. Statistical significance at the 10%, 5%, and 1% levels is denoted by *, **, and ***.

Table 8 reports the results. Estimates based on all citations and applicant-added citations are similar in magnitude and statistically significant within each year. Including examiner-added citations therefore does not materially affect the measurement of absorption intensity or the conclusions.

5. External validity

Having established that absorption intensity amplifies firms' innovation response to AIPA, we next examine whether absorption intensity also predicts outcomes beyond the AIPA setting. Economic theory and prior research suggest that firms capable of assimilating external knowledge should outperform others over time: patents can enable new products that raise sales and investment (Kogan et al., 2017), improve production efficiency and automation (Bena et al., 2022), deter competitors and strengthen market power (Ziedonis, 2004), and generate licensing revenue (Arora et al., 2004). If absorption intensity reflects a firm's underlying capability to process and apply external ideas, it should also be associated with superior innovation and firm growth even in the absence of major policy shocks.

We first test whether absorption intensity is correlated with future patent output. For this analysis, we use a panel spanning 1975 through 2019. The first row in Panel A of Table 9 shows that absorption intensity is strongly associated with patent output up to five years ahead, indicating that firms with greater absorption capability sustain higher innovation rates well beyond the AIPA window.

We next turn to patent quality. Although our AIPA analysis shows no differential intensive-margin response in patent characteristics among high-absorption, highly treated firms, absorption capability should nonetheless be associated with higher-quality innovation more generally if it reflects a firm's ability to identify and integrate external knowledge. The remaining rows in Panel A show that higher absorption intensity is indeed associated with patents that are more novel (more original and general), less defensive (lower self-citation share), broader in scope (more claims and fewer words), more scientifically impactful (more forward citations), and more economically valuable (higher economic value, higher renewal rates, and lower technological obsolescence).

Finally, we test whether absorption intensity correlates with broader measures of firm performance. Following Kogan et al. (2017) and Ma (2025), we analyze growth in profit, labor, physical capital, output, and total factor productivity (TFP). We extend this framework by including growth in R&D capital – a novel dimension reflecting the rising importance of intangible investment in modern firms (Haskel and Westlake, 2018). Panel B of Table 9 shows that absorption intensity is most strongly associated with growth in R&D capital, consistent with the idea that innovation success facilitates reinvestment in research capacity (Hsu and Ziedonis, 2013). We also find significant positive correlations with profit and employment growth, while associations with output and physical capital are smaller. The lack of a robust relation with TFP aligns with evidence that patents affect appropriability and market outcomes more directly than measured productivity (Scherer, 1982).

Although the correlations in Table 9 are not causal, the consistent associations with innovation output and firm performance indicate that absorption intensity is systematically related to firm growth.

6. Antecedents of absorption intensity

Finally, we examine firm-level determinants of absorption intensity. The capability to recognize, assimilate, and apply external information depends on organizational structures that support innovation, external linkages that provide access to new ideas, and the human capital that enables their effective use (Cohen and Levinthal, 1989, 1990). Guided by these insights, we relate absorption intensity to observable firm characteristics in each of these domains.

Organizational factors: Organizational design shapes how effectively firms transmit and integrate knowledge. Internal information flows and communication structures determine how external ideas are recognized and incorporated into firm strategy (Cohen and Levinthal, 1990). Communication between senior managers and inventors facilitates this process, and spatial proximity further supports interaction (Akerlof, 1997). Firms whose inventors are located closer to headquarters should therefore exhibit higher absorption intensity. We measure proximity using inventor distance – the average physical distance between inventors and firm headquarters – computed from patent records over the five-year window ending five years prior. Headquarters locations are drawn from Gao et al. (2021). The sample begins in 2009 due to data availability. The first column in Panel A of Table 10 shows that absorption intensity is strongly negatively correlated with inventor distance, consistent with the idea that spatial proximity enhances a firm's capability to absorb external knowledge.

A complementary aspect of organizational design is culture. Culture shapes how employees respond to new information and uncertainty (Davenport et al., 1998; Harrington and Guimaraes, 2005), and cultures that tolerate failure and reward exploration encourage engagement with external ideas (Manso, 2011; Tian and Wang, 2014). Firms with stronger innovation cultures should therefore display higher absorption intensity. We measure innovation culture as the natural log of the

Table 9
External validity.

Panel A: Patent novelty, scope and value					
	t+1	t+2	t+3	t+4	t+5
Output	0.13**	0.19***	0.23***	0.27***	0.30***
	(2.47)	(2.97)	(3.16)	(3.33)	(3.42)
Originality	0.13***	0.12***	0.11***	0.09***	0.08***
	(13.66)	(12.41)	(9.79)	(9.31)	(7.70)
Generality	0.14***	0.13***	0.11***	0.11***	0.09***
	(10.79)	(8.75)	(7.47)	(7.25)	(6.04)
Self-cites share	−0.27***	−0.24***	−0.22***	−0.20***	−0.19***
	(−11.38)	(−10.58)	(−9.07)	(−9.15)	(−7.90)
Claims	0.06***	0.05***	0.04***	0.03***	0.02***
	(6.71)	(6.93)	(5.46)	(4.06)	(2.69)
Words	−0.01**	−0.02**	−0.02**	−0.01*	−0.01
	(−2.03)	(−2.52)	(−2.44)	(−1.66)	(−1.49)
Forward citations	0.31***	0.28***	0.26***	0.23***	0.19***
	(9.40)	(8.10)	(7.75)	(6.25)	(4.91)
Economic value	0.23***	0.25***	0.25***	0.24***	0.21***
	(5.79)	(5.89)	(5.19)	(4.63)	(4.14)
Renewal share	0.05***	0.05***	0.05***	0.04**	0.04**
	(3.33)	(2.89)	(2.80)	(2.17)	(2.15)
Technological obsolescence	−0.07***	−0.08***	−0.07***	−0.08***	−0.07***
	(−6.31)	(−6.04)	(−6.27)	(−7.22)	(−6.08)

Panel B: Firm growth					
	t+1	t+2	t+3	t+4	t+5
R&D capital	0.006***	0.014***	0.023***	0.034***	0.043***
	(3.46)	(3.83)	(4.28)	(4.80)	(4.62)
Profit	0.006**	0.013**	0.017**	0.023***	0.029***
	(2.23)	(2.54)	(2.57)	(2.79)	(3.07)
Labor	0.003*	0.007*	0.011**	0.016**	0.021**
	(1.94)	(1.94)	(2.32)	(2.44)	(2.51)
Output	0.003	0.006	0.010*	0.014*	0.018*
	(1.22)	(1.25)	(1.77)	(1.83)	(1.86)
Physical capital	0.001	0.003	0.007	0.011	0.018*
	(0.70)	(0.65)	(1.20)	(1.51)	(1.94)
TFP	−0.002	−0.002	−0.001	−0.002	0.002
	(−0.54)	(−0.52)	(−0.27)	(−0.38)	(0.23)

This table presents long-run correlations between absorption intensity and patent output or firm growth. In Panel A, we focus on patent characteristics. In Panel B, we focus on measures of firm growth, which we define in log changes. Controls include the variables reported in Table 1. When growth in output or TFP is the dependent variable, we additionally control for their respective levels. Besides leverage, all independent variables are logged. See Appendix A for variable definitions. The sample is from 1975 through 2019. We estimate Poisson regressions in Panel A (with the exception of technological obsolescence, which is estimated using OLS) and OLS regressions in Panel B. Standard errors are double clustered at the firm and year levels. Statistical significance at the 10%, 5%, and 1% levels is denoted by *, **, and ***.

corporate cultural value of innovation from Li et al. (2021). The sample begins in 2007. The second column in Panel A shows that absorption intensity is positively and significantly correlated with innovation culture, consistent with the view that an open, experimental environment improves firms' capability to integrate external knowledge.

Local research ecosystem: A firm's capability to absorb external ideas depends not only on its internal organization but also on the local research ecosystem. Universities are often central to the local research ecosystem: they generate frontier science (Furman and MacGarvie, 2007), train researchers who contribute to firms' human and social capital (Murray, 2004), and anchor networks that facilitate collaboration (Owen-Smith and Powell, 2004). Firms located near high-quality academic institutions should therefore exhibit higher absorption intensity. To test this prediction, we measure local research quality using the SCImago Institution Rankings, which combine research output (50%), innovation (30%), and societal impact (20%). For each firm, we compute academic research rank as the average rank of universities within 100 miles of headquarters, measured five years prior. The sample begins in 2014. The third column in Panel A shows that absorption intensity is positively and significantly related to nearby university research quality, consistent with the idea that a strong local research ecosystem strengthens firms' absorption capability.[15]

Human capital: Firms' human capital underpins their capability to recognize and apply external knowledge (Zahra and George, 2002). Within firms, inventors represent a critical component of this human capital, serving as the primary channel through which new ideas are generated and integrated. Firms with higher quality inventors should therefore exhibit greater absorption intensity. We measure inventor quality using inventor citation rank, defined as the average citation rank of all inventors listed on patents granted in the five years ending five years prior. Inventors are identified using the USPTO inventor disambiguation file, and rankings are computed across all U.S. inventors, not only those employed by sample firms. The sample begins in 1980. The last column in Panel A of Table 10 shows that absorption intensity is positively correlated with inventor citation rank, consistent with the view that higher quality inventors raise firms' capability to process external knowledge.

Another dimension of human capital is the breadth of employees' prior knowledge, which facilitates the recognition and assimilation of new information (Cohen and Levinthal, 1990). Formal education expands this knowledge base, implying that firms with more educated employees should exhibit higher absorption intensity. We test this prediction using data from the World Management Survey (WMS) on education levels in medium-sized manufacturing firms across countries (Bloom and Van Reenen, 2007; Bloom et al., 2012).[16] We focus on the share of managers and non-managers with college degrees, both lagged five years. After matching to our sample, fewer than 65 firms remain, so we emphasize correlations rather than regression estimates.

[15] Results are similar when using 50- or 150-mile radii around firm headquarters.

[16] We thank Nicholas Bloom and the WMS team for providing access to the data.

Table 10
Antecedents of absorption intensity.

Panel A: Absorption intensity and its antecedents

	(1)	(2)	(3)	(4)
Inventor distance$_{t-5}$	−0.045***			
	(−3.43)			
Innovation culture$_{t-5}$		0.067**		
		(2.34)		
Academic research rank$_{t-5}$			0.243**	
			(2.99)	
Inventor citation rank$_{t-5}$				0.190***
				(4.25)
Controls	Yes	Yes	Yes	Yes
Industry × Year FE	Yes	Yes	Yes	Yes
Adjusted R^2	29.9%	31.5%	18.2%	35.7%
Observations	5866	8760	4774	21,165

Panel B: Absorption intensity versus its antecedents

	(1)	(2)	(3)	(4)
Absorption intensity	0.15**	0.18***	0.08	0.14**
	(2.46)	(3.21)	(1.24)	(2.17)
Inventor distance	−0.04			
	(−1.41)			
Innovation (culture)		0.01		
		(0.19)		
Academic research rank			0.05	
			(0.48)	
Inventor citation rank				−0.002
				(−0.96)
Controls	Yes	Yes	Yes	Yes
Industry × Year FE	Yes	Yes	Yes	Yes
Pseudo R^2	91.7%	91.5%	92.7%	91.8%
Observations	9861	14,757	9949	35,608

This table presents results from regressions that focus on potential antecedents of absorption intensity. Panel A presents results from OLS regressions of absorption intensity on its potential antecedents, which are lagged five years. Panel B presents results from Poisson regressions of innovation output on absorption intensity and its potential antecedents. Controls include the variables reported in Table 1. Besides leverage, all independent variables are logged. See Appendix A for variable definitions. Standard errors are double clustered at the firm and year levels. Statistical significance at the 10%, 5%, and 1% levels is denoted by *, **, and ***.

Absorption intensity is positively correlated with educational attainment at both levels: 18% for managers and 30% for non-managers. Since education varies systematically across industries, we remove industry-level means from each measure. The correlations then rise to 63% for managers and remain sizable at 32% for non-managers. These patterns are consistent with the human capital interpretation, though the small sample size warrants caution.[17]

Antecedents and innovation output: A natural question is whether any of these firm-level determinants of absorption intensity also predict innovation output directly. Panel B of Table 10 shows they do not: none of the antecedent variables are significantly related to one-year ahead patent output, while absorption intensity remains positive and highly significant in most specifications.

7. Conclusion

In this paper, we develop a new, outcome-based measure of firms' capability to absorb external knowledge – absorption intensity – that captures how effectively firms incorporate external ideas into their own inventions. Exploiting the American Inventors Protection Act of 1999 as an exogenous improvement in the external information environment, we show that firms with higher absorption intensity experience substantially larger increases in innovation output when exposure to newly disclosed patent information is greater. This result establishes that heterogeneity in firms' capacity to process and use external information – rather than differences in R&D spending or technological opportunity alone – is an important determinant of how knowledge flows translate into innovation.

Beyond the causal evidence, our analysis shows that absorption intensity captures a persistent and economically meaningful dimension of firm behavior. Firms with higher absorption intensity consistently generate more and higher value patents and exhibit stronger subsequent growth in R&D, profits, and employment. At the same time, absorption intensity reflects the combined influence of multiple underlying determinants in a single, observable outcome measure, providing a tractable empirical proxy for firms' absorptive capacity.

Our findings have important implications for research in innovation and corporate finance. Differences in absorptive capacity shape firms' responses to changes in the information environment, affect the returns to intangible investment, and help explain persistent heterogeneity in innovation and growth among otherwise similar firms. Absorption intensity offers a practical tool for future research on how information flows influence real investment decisions, the organization of firm boundaries, and the valuation consequences of strategies that rely on the effective transfer of external knowledge.

CRediT authorship contribution statement

Michael Woeppel: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Resources, Validation, Visualization, Writing – original draft, Writing – review & editing. **M. Deniz Yavuz:** Conceptualization, Formal analysis, Investigation, Methodology, Project administration, Resources, Validation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors have no conflict of interest.

Appendix A. Variable definitions

Control variables	
Firm age	Number of years in the Compustat database.
Labor	Number of employees.
Leverage	Sum of current debt and long-term debt divided by total assets.
Market-to-book	Market value of equity plus preferred stock and total debt minus deferred taxes all divided by total assets.
Patent count	Natural logarithm of the total number of active patents.
Physical capital	Gross property, plant, and equipment adjusted for the age of capital, deflated by the NIPA price index for equipment (İmrohoroğlu and Tüzel, 2014).
Profit	Sales minus cost of goods sold, deflated by the CPI.
R&D capital	Cumulated and depreciated stock of R&D expense, deflated by the CPI (Peters and Taylor, 2017).
Tangibility	Net property, plant, and equipment divided by total assets.
R&D-based proxies	
R&D expense	Research and development expense, deflated by the CPI.
R&D dummy	Indicator of whether the firm reports positive R&D expense.

[17] We also find that absorption intensity is weakly correlated (3%) with the overall WMS management score, which is consistent with the broad scope of the WMS management questions.

R&D/assets	Ratio of R&D expense to book assets.
R&D/labor	Ratio of the three-year sum of R&D expense to the number of employees (Tsai, 2009).
R&D/sales	Ratio of R&D expense to sales.
Lagged inputs	
Lagged capital expenditures	Natural logarithm of the previous year's capital expenditures, deflated by the CPI.
Lagged labor	Natural logarithm of the previous year's number of employees.
Lagged R&D expense	Natural logarithm of previous year's research and development expense, deflated by the CPI.
Patent characteristics	
Breakthrough	Number of patents classified as breakthrough patents (i.e., patents that are textually different from past patents but textually similar to future patents) (Kogan et al., 2021).
Claims	Number of independent claims per patent (Marco et al., 2019).
Core	Number of patents in CPC classes corresponding to the firm's top three CPC classes in its 2000 patent portfolio (Akcigit and Kerr, 2018).
Economic value	Dollar value of innovation measured using abnormal stock returns around patent grant announcements, deflated by CPI (Kogan et al., 2017).
Examiner share of self-cites	Examiner-added self-citations divided by total firm self-citations (Sampat, 2010).
Exploitative	Number of patents for which at least 80% of citations are to patents previously cited by the firm (Benner and Tushman, 2002).
Exploratory	Number of patents for which at least 80% of citations are to patents not previously cited by the firm (Benner and Tushman, 2002).
Forward citations	Number of forward external citations per patent within ten years of publication (Trajtenberg, 1990).
Generality	One minus the Herfindahl index of a patent's forward citations across technological fields (Trajtenberg et al., 1997).
Non-breakthrough	Number of patents minus the number of breakthrough patents.
Non-core	Number of patents minus the number of core patents.
Originality	One minus the Herfindahl index of a patent's backward citations across technological fields (Trajtenberg et al., 1997).
Relative spillover	Ratio of rivals' pre-AIPA grant lag to the firm's own pre-AIPA grant lag, computed using patents over the past five years (Kim and Valentine, 2021).
Renewal share	Share of patents that are subsequently renewed at the 12-year maintenance-fee stage (Sampat, 2010).
Self-cites share	Share of citations to patents by the same applicant, averaged across patents (Hall et al., 2001).
Technological obsolescence	The rate at which a firm's technology stock loses relevance over time, measured by the decay in external citations to the patents cited by its patent portfolio (Ma, 2025).
Words	Number of words in the shortest independent claim per patent (Marco et al., 2019).
Other variables	
Academic research rank	Average SCImago Institution Ranking of academic institutions located within 100 miles of the firm's headquarters. The SCImago ranking aggregates research output (50%), innovation (30%), and societal impact (20%).
Innovation culture	Natural log of the corporate cultural value of innovation (Li et al., 2021).
Inventor citation rank	Average citation rank of inventors on patents granted in the past five years.
Inventor distance	Average geographic distance between firm headquarters and inventors on patents granted in the past five years.
Output	Sales plus change in inventory, deflated by the CPI (Kogan et al., 2017).
TFP	Firm-level total factor productivity, measured as output net of capital and labor inputs (İmrohoroğlu and Tüzel, 2014).

Unless otherwise noted, all variables are winsorized at the 1% level in each tail annually.

Appendix B. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.jfineco.2026.104322.

Data availability

Absorption intensity replication code and data) (Mendeley Data)

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